Response to AS2: Homework 2 “Make presentation quality graphics”

Ethan Suresh*
MIT Department of Physics
(Dated: February 21, 2026)

We analyze a dataset of gravitational acceleration measurements obtained from simple pendulum experiments conducted across students from multiple past J-Lab classes. After rejecting outliers using an interquartile-range criterion (data points must be within 10 IQR from the median), we fit a Gaussian distribution to the distribution of measured g values using a chi-squared regression algorithm. We find a value of $g = 9.794 \pm 0.007$ m/s^2, consistent with the accepted value of $g = 9.804$ m/s^2 for the Boston area to within 1.42σ. The fit yields a high $\chi^2/\mathrm{dof} = 168.7/36$ as well as an extremely low probability of fit $p = 0.00$, primarily due to long non-Gaussian tails. This indicating that the Gaussian distribution is a poor fit for this dataset.

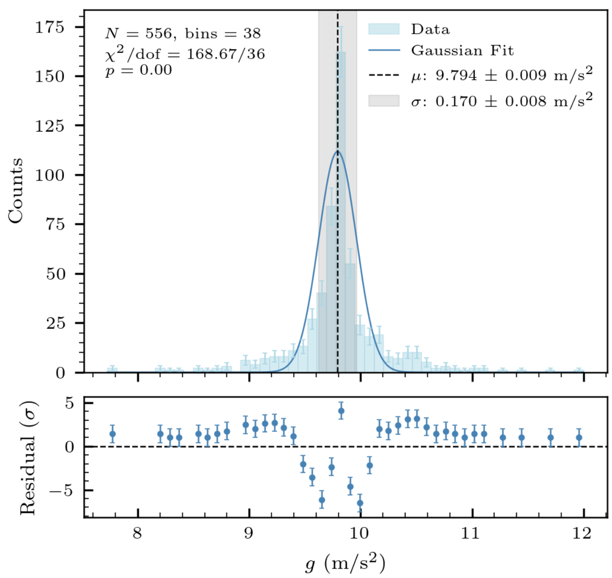

FIG. 1. A histogram distribution of gravitational acceleration g measurements ($N = 556$ trials after outlier rejection). The uncertainty is each bin is taken to be the square root of the number of counts in that bin, following a Poisson distribution. The histogram is fit to a Gaussian (solid curve), using a Chi-squared regression algorithm. The regression yielded fit parameters $\mu = 9.794 \pm 0.009$m/s^2 (dashed vertical line) and $\sigma = 0.170 \pm 0.008$m/s^2 (shaded band). From these parameters, the uncertainty in the mean is calculated as $\sigma_\mu = \sigma/\sqrt{N} = 0.007$m/s^2, The lower panel shows the normalized residuals, which show a non-random distribution of errors that are often further than one standard error from the prediction, resulting in a large. In particular, the tails of the Gaussian prediction systematically underestimate the data while the peak behavior overestimates the data. Thus, the probability of fit p is extremely small.

ACKNOWLEDGMENTS

The dataset for this document was generated by students in previous years of Junior Lab and organized by Junior Lab staff. Additionally, the template for this document was adapted from the American Physical Society’s REVTeK-4.1 template by Junior Lab staff, and the code used to perform the Chi-Squared fit was adapted from a template created by Junior Lab Staff. Finally, I am grateful to the Junior Lab staff for their advice in preparing this analysis.

* easuresh@mit.edu; http://web.mit.edu/8.13/

[1] M. D. o. P. Helena Foundation Junior Laboratory Staff, “Junior Lab 8.13 Syllabus,” (2026).
[2] M. D. o. P. Helena Foundation Junior Laboratory Staff,

"Laboratory Safety and Regulations in Junior Lab," (2023).

[3] M. D. o. P. Helena Foundation Junior Laboratory Staff, "Ethics in Science and Education," (2023).

[4] MIT Department of Physics, "Community Values," .

[5] M. D. o. P. Helena Foundation Junior Laboratory Staff, "Requirements for Experimental Notebooks," (2023).